2D SPDE FINAL RESULTS: FNO vs Vanilla PINN vs Spectral-PINN

SPATIAL PSD ERROR (sp_err) — Lower is Better

Channel	FNO	Vanilla PINN	Spectral-PINN	Improvement (Spectral vs Vanilla)
z (Correlation)	4.6e-4	1.21 ± 0.13	0.34 ± 0.11	3.6x ✓
y (Volatility)	2.2e-4	1.13 ± 0.12	0.24 ± 0.06	4.7x ✓
p (Density)	5.6e-4	0.86 ± 0.09	0.21 ± 0.09	4.1x ✓

TEMPORAL PSD ERROR (tp_err) — Lower is Better

z (Correlation)	2.6e-3	1.68 ± 0.26	0.39 ± 0.10	4.4x ✓
y (Volatility)	9.0e-4	0.87 ± 0.26	0.26 ± 0.08	3.3x ✓
p (Density)	8.9e-4	0.53 ± 0.33	0.014 ± 0.008	38.7x ✓✓✓

VARIANCE RATIO (target = 1.0) — Closer to 1.0 is Better

z (Correlation)	0.978	0.19 ± 0.10	0.825 ± 0.072	4.3x better (target match)
y (Volatility)	1.101	0.75 ± 0.43	0.795 ± 0.443	Near target
p (Density)	0.962	0.45 ± 0.21	0.792 ± 0.095	1.8x better

KEY INSIGHTS

- **FNO PERFORMANCE:** 1000x better than Vanilla PINN. Near-perfect variance matching (0.96-1.1).
  Fast inference (~100x faster than PINN). Single deterministic model.
  Cost: Required parameter shrinking (53M→8.4M), torch.compile debugging.

- **VANILLA PINN FAILURE:** Catastrophic failure on stochastic PDEs. Residual loss blind to spectral structure.
  Creates smooth, unphysical solutions. Variance 4-5x too small. sp_err ~1.2.
  Root cause: Minimizes PDE residual but misses distributional properties.

- **SPECTRAL-PINN SUCCESS:** Fixes vanilla PINN via 3 novel distributional loss terms on 32×32×128 sub-grid:
  - L_spec_xy: Spatial PSD matching
  - L_spec_t: Temporal PSD matching
  - L_var: Variance ratio matching (weight=50.0, dominates early training)

  Results: 3-4x error reduction, variance ratio 0.82 (near target 1.0).
  Still 1000x behind FNO but VIABLE for uncertainty quantification + physics.

- **ENSEMBLE VARIANCE:** 8 Spectral-PINN seeds × 4 test cases show consistent performance.
  Std of metrics: 0.06-0.44 across all channels.
  Enables principled uncertainty quantification for downstream applications.

RECOMMENDATION

Use FNO for: Maximum accuracy, speed, deterministic predictions. Production inference.
Use Spectral-PINN: Uncertainty quantification, ensemble variance, physics constraint enforcement.
Academic/research applications, where ensemble + distributional properties matter.